

Tuesday, Feb 24

- Missed quiz policy

1. Reach out before class
2. Take quiz at first OH after

Goal: Check understanding

Incentivize class attendance

- Late assignment policy

- ↳ 24 hours late window (life happens)
- ↳ hard deadline after that

Plan

- Logistic Regression

- Pytorch demo

Optimization

exact or gradient descent

Both need the gradient, so...

lemma: $\nabla_{\omega} \mathcal{L}(\omega) = X^T (\sigma(X\omega) - y)$

$$z = \langle \omega, x \rangle, \quad \hat{y} = \sigma(z)$$

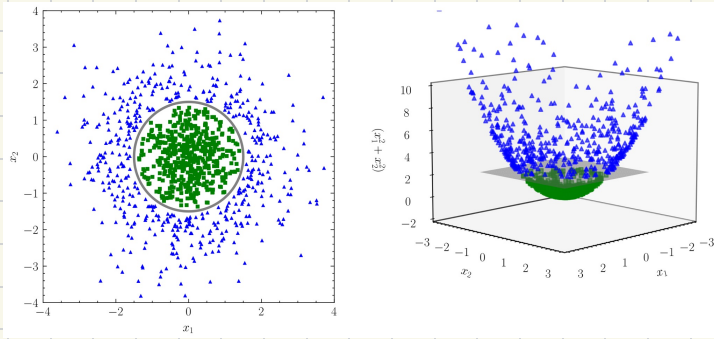
Fact 1: $\frac{\partial \sigma(z)}{\partial z} = \sigma(z)(1 - \sigma(z)) = \hat{y}(1 - \hat{y})$

$$\ell = -y \log \hat{y} - (1-y) \log (1-\hat{y})$$

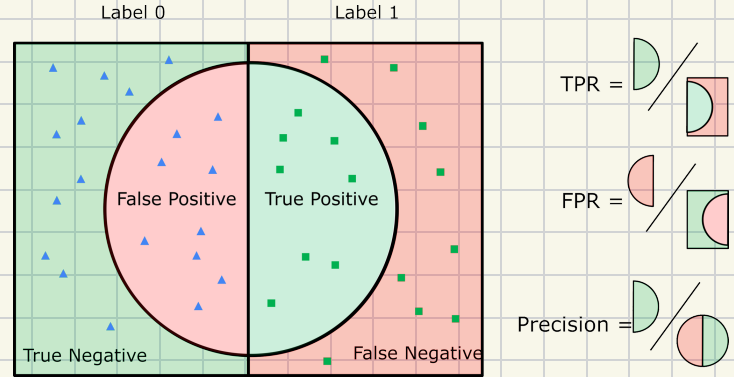
Fact 2: $\frac{\partial \ell}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$

$$\begin{aligned} \nabla_{\omega} \mathcal{L}(\omega) &= \sum_{i=1}^n \frac{\partial \ell^{(i)}}{\partial \hat{y}^{(i)}} \frac{\partial \hat{y}^{(i)}}{\partial z^{(i)}} \nabla_{\omega} z^{(i)} \\ &= \dots \end{aligned}$$

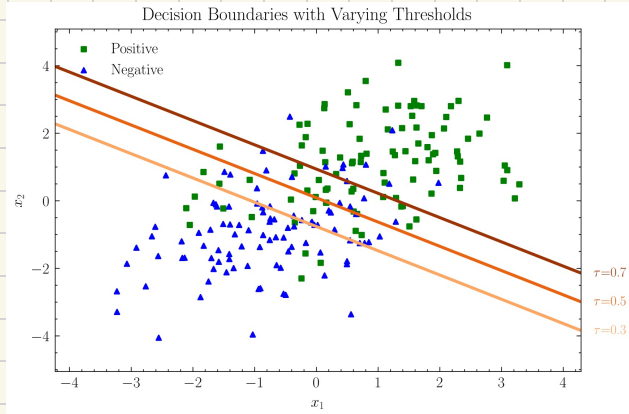
Non-linear Transformations



Measuring Error



Varying Threshold



Output 1 if $\sigma(\langle w, x \rangle) > \frac{1}{2}$ $\nearrow \tau$

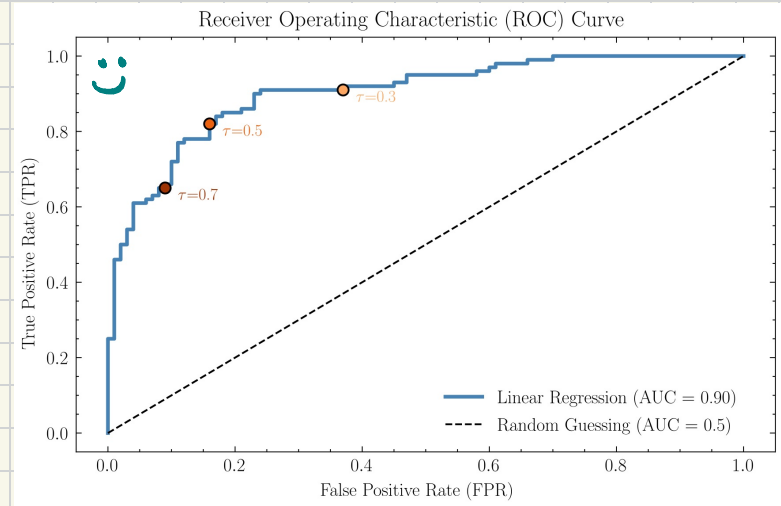
Output 0 else

Decreasing τ :

↑ TPR

↑ FPR

Receiver Operating Characteristic Curve



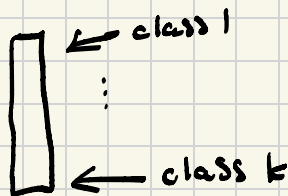
Multiple Classes

$$f: \mathbb{R}^d \rightarrow \mathbb{R}^K$$

$$W \in \mathbb{R}^{K \times d}$$

$$Wx$$

$$K \times d \quad d \times 1$$



Goal: Convert to probability distribution

1. Between 0 and 1

2. Sum to 1

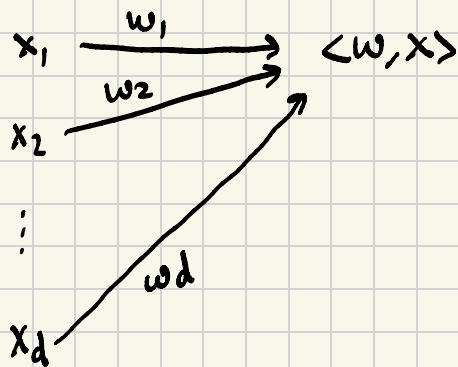
$$z \in \mathbb{R}^K$$

$$\text{softmax}(z) = \frac{1}{\sum_{j=1}^K e^{z_j}} \begin{bmatrix} e^{z_1} \\ e^{z_2} \\ \vdots \\ e^{z_K} \end{bmatrix}$$

$$f(x) = \text{softmax}(Wx)$$

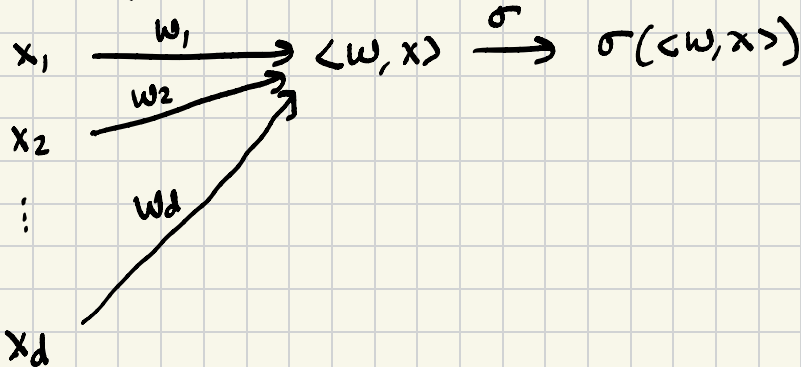
$$\mathcal{L}(W) = - \sum_{i=1}^n \sum_{j=1}^K \mathbb{I}[y^{(i)} = j] \log(f_j(x^{(i)}))$$

Linear Regression $f: \mathbb{R}^d \rightarrow \mathbb{R}$



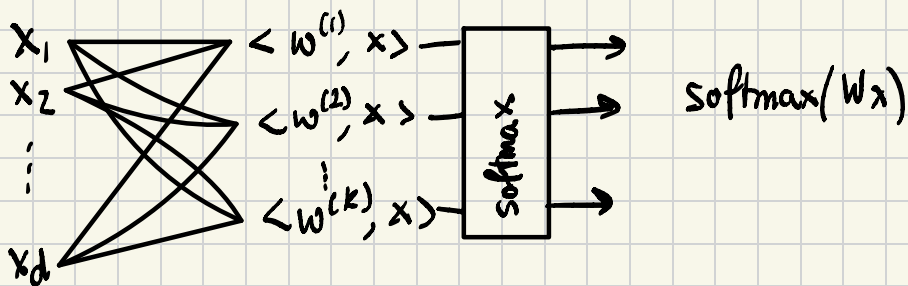
$$\mathcal{L}(w) = \frac{1}{n} \sum_{i=1}^n (f(x^{(i)}) - y^{(i)})^2$$

Binary Logistic Regression $f: \mathbb{R}^d \rightarrow (0,1)$



$$\mathcal{L}(w) = - \sum_{i=1}^n y^{(i)} \log(f(x^{(i)})) + (1 - y^{(i)}) \log(1 - f(x^{(i)}))$$

Multi-class Logistic Regression $f: \mathbb{R}^d \rightarrow (0,1)^k$



$$\mathcal{L}(w) = - \sum_{i=1}^n \sum_{j=1}^k \mathbb{1}[\tilde{y}^{(i)} = j] \log f_j(x^{(i)})$$